```
import matplotlib.pyplot as plt
```

Given daily temperature data for 30 days, which chart will you use and why? linechart

You are comparing sales of 5 different products in a month. Which chart is most suitable?

Bar Chart

Marks of 100 students are given. Which visualization will best show how marks are distributed?

Histogram

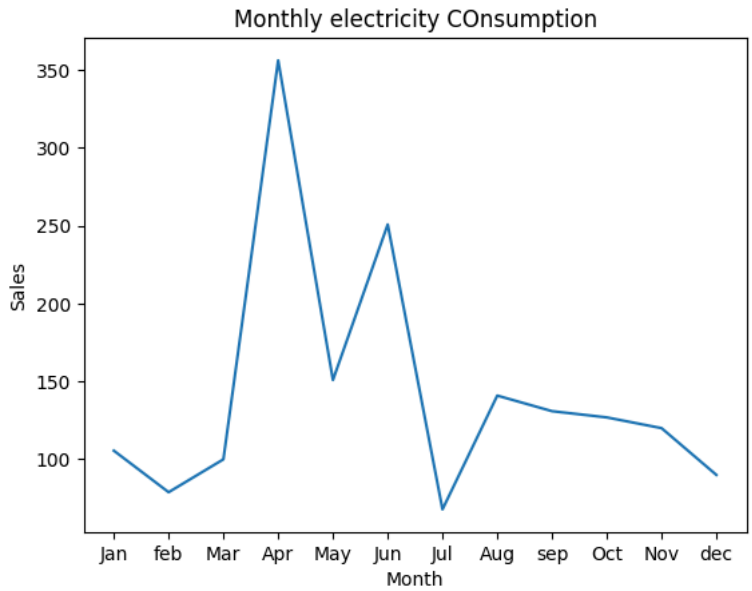
You have yearly revenue data for the last 10 years. Which chart will clearly show the trend?

Line Chart

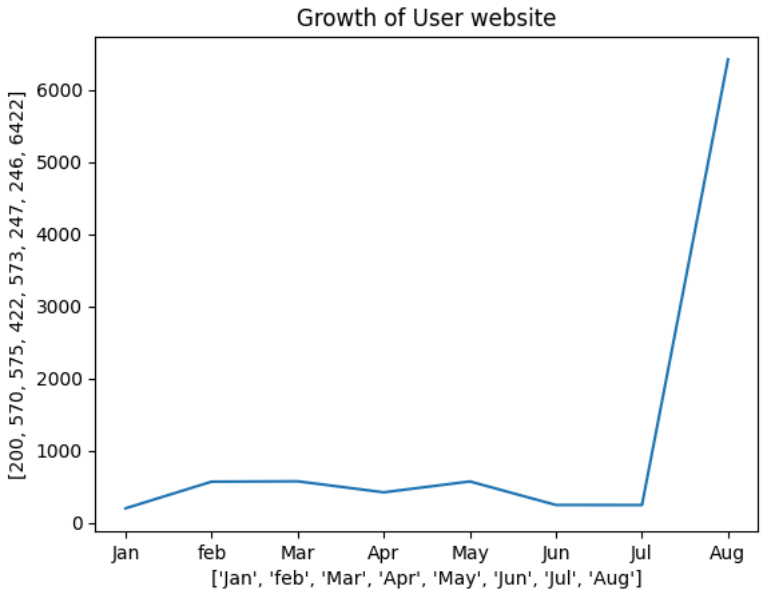
Website response times are stored in a list. Which chart will help analyze system performance?

Histogram

```
#Plot a line chart showing monthly electricity consumption for 12 months.
Month=['Jan','feb','Mar','Apr','May','Jun','Jul','Aug','sep','Oct','Nov','dec']
Units=[105.56,78.90,100,356,150.89,250.70,67.90,140.90,130.90,126.90,120,90]
plt.plot(Month,Units)
plt.xlabel("Month")
plt.ylabel("Sales")
plt.title("Monthly electricity COnsumption")
plt.show()
```

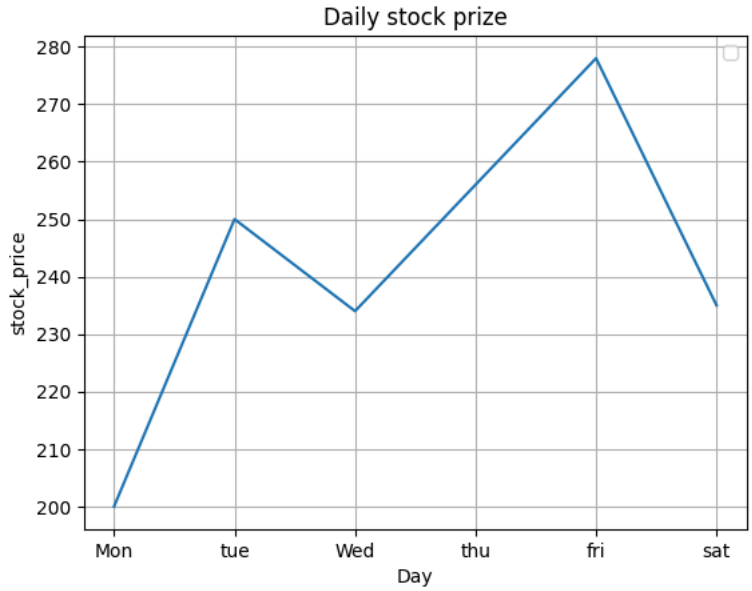



```
#Create a line chart to show the growth of website users over 8 months.
Month=['Jan','feb','Mar','Apr','May','Jun','Jul','Aug']
Growth=[200,570,575,422,573,247,246,6422]
plt.plot(Month,Growth)
plt.xlabel(Month)
plt.ylabel(Growth)
plt.title("Growth of User website")
plt.show()
```



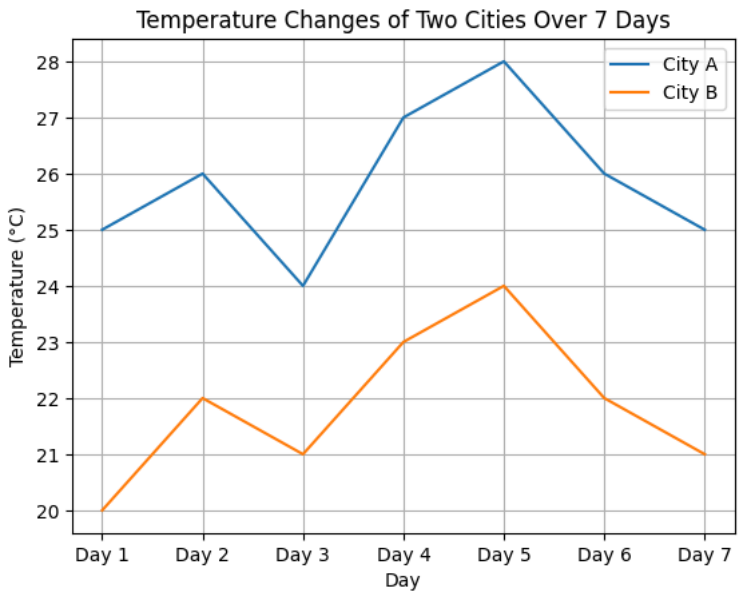

```
#Draw a line chart for daily stock prices and identify whether the trend is upward or downward.
Day=["Mon","tue",'Wed','thu','fri','sat']
stock_price=[200,250,234,256,278,235,]
plt.plot(Day,stock_price)
plt.xlabel("Day")
plt.ylabel("stock_price")
plt.title("Daily stock prize")
plt.grid(True)
plt.show()
```

```
/tmp/ipython-input-3267755238.py:8: UserWarning: No artists with labels found to put in legend.  Note that artists whose lab
  plt.legend()
```

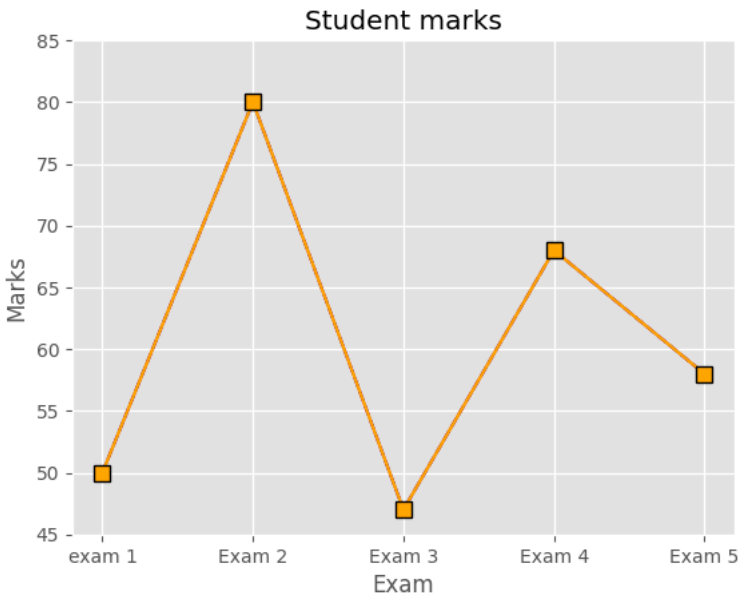



```
# Data for 7 days
Days = ['Day 1', 'Day 2', 'Day 3', 'Day 4', 'Day 5', 'Day 6', 'Day 7']

# Temperature data for two cities
CityA_Temps = [25, 26, 24, 27, 28, 26, 25]
CityB_Temps = [20, 22, 21, 23, 24, 22, 21]

# Plotting the line chart
plt.plot(Days, CityA_Temps, label='City A')
plt.plot(Days, CityB_Temps, label='City B')

# Adding labels and title
plt.xlabel("Day")
plt.ylabel("Temperature (°C)")
plt.title("Temperature Changes of Two Cities Over 7 Days")
plt.legend()
```

```
plt.grid(True)
plt.show()
```

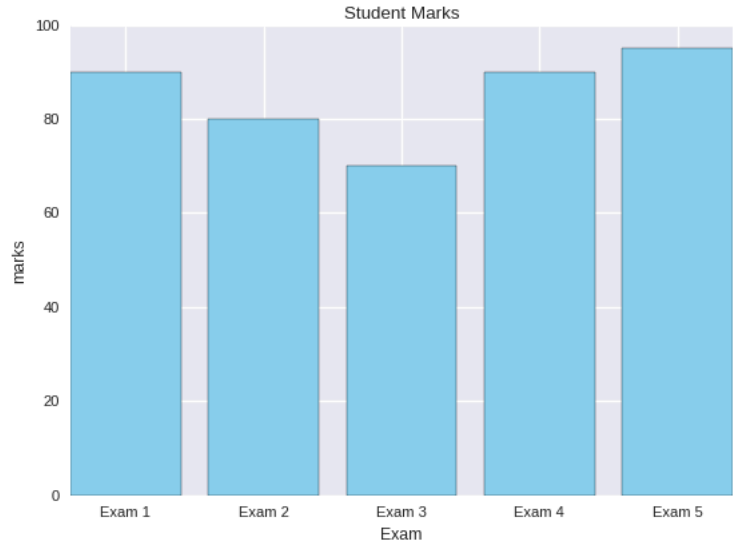



```
#Create a line chart showing progress of a student’s marks across 5 exams.

student_marks=[50,80,47,68,58]
exam=["exam 1","Exam 2","Exam 3","Exam 4","Exam 5"]

plt.plot(exam,student_marks)
plt.xlabel("Exam")
plt.ylabel("Marks")
plt.title("Student marks")
plt.plot(exam, student_marks, marker='o', linestyle='--', color='purple')
plt.plot(exam, student_marks, marker='s', markersize=8, color='orange')
plt.style.use('classic')   # other options: 'ggplot', 'classic', 'dark_background'

plt.grid(True)
plt.show()
```

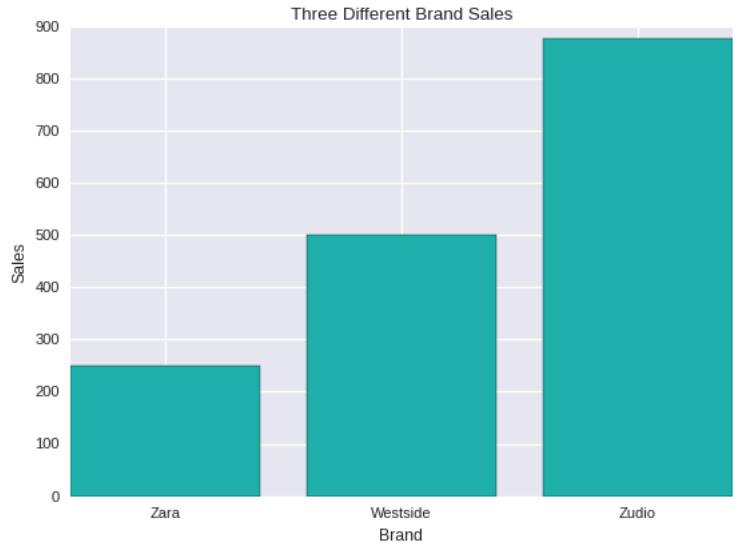



```
#Create a bar chart showing marks scored by 5 students in an exam.


marks=[90,80,70,90,95]
Exam=["Exam 1","Exam 2","Exam 3","Exam 4","Exam 5"]

colors=['skyblue']
plt.bar(Exam,marks,color=colors)
plt.xlabel("Exam")
plt.ylabel("marks")
plt.title("Student Marks")
plt.style.use('seaborn-v0_8')
```

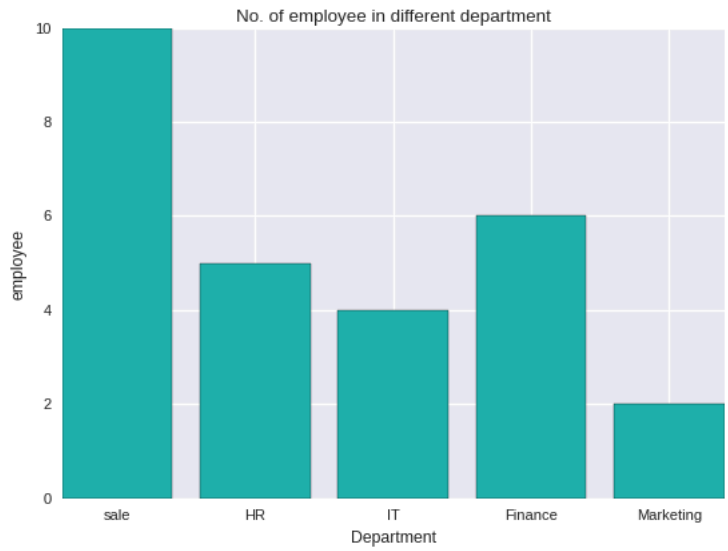
```
plt.show()
```



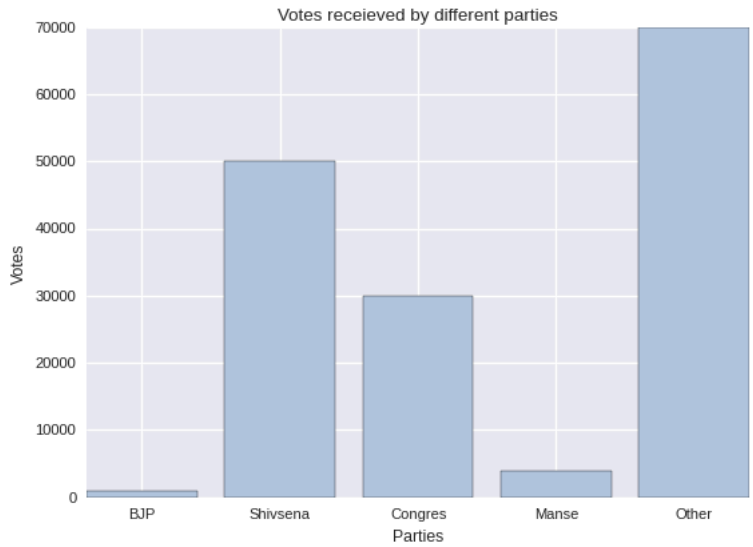

```
#Plot a bar chart comparing sales of three different brands.

sales=[250,500,876]
brands=["Zara","Westside","Zudio"]
plt.bar(brands,sales,color=["lightseagreen"])
plt.xlabel("Brand")
plt.ylabel("Sales")
plt.title("Three Different Brand Sales")
plt.style.use('seaborn-v0_8')
plt.show()
```



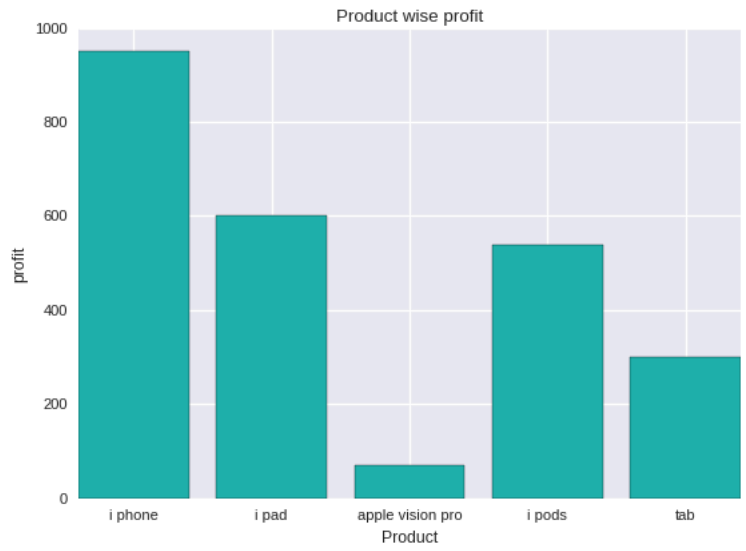

```
#Draw a bar chart to represent number of employees in different departments.

employee=[10,5,4,6,2]
Department=['sale','HR','IT','Finance','Marketing']
plt.bar(Department,employee,color=["lightseagreen"])
plt.xlabel("Department")
plt.ylabel("employee")
plt.title("No. of employee in different department")
plt.show()
```



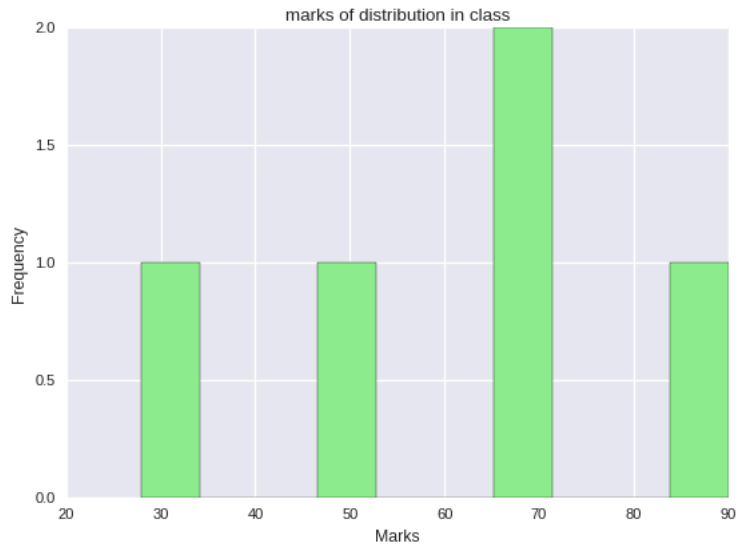

```
#Create a bar chart to compare votes received by different political parties.

vote=[1000,50000,30000,4000,70000]
parties=["BJP","Shivsena","Congres","Manse", "Other"]
colors=["lightsteelblue"]
plt.bar(parties,vote,color=colors)
plt.xlabel("Parties")
plt.ylabel("Votes")
plt.title("Votes receieved by different parties")
plt.show()
```




```
#Plot a bar chart showing product-wise profit in a company.

#Company=Apple
product=["i phone","i pad","apple vision pro","i pods","tab"]
profit=[950,600,70,540,300]
plt.bar(product,profit,color=["lightseagreen"])
plt.xlabel("Product")
plt.ylabel("profit")
plt.title("Product wise profit")
plt.show()
```

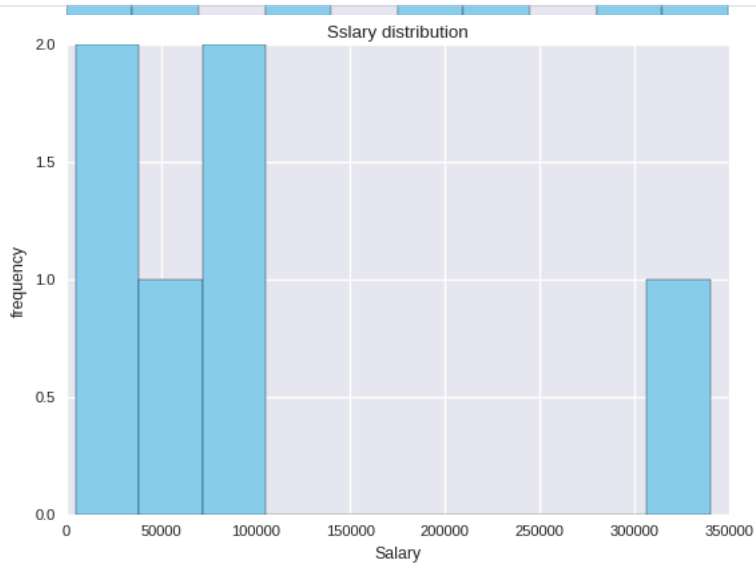



```
#Create a histogram to show the distribution of marks of students in a class.

marks=[50,70,28,70,90]
plt.hist(marks,bins=10,color=["lightgreen"])
plt.xlabel("Marks")
plt.ylabel("Frequency")
plt.title("marks of distribution in class")
plt.show()
```

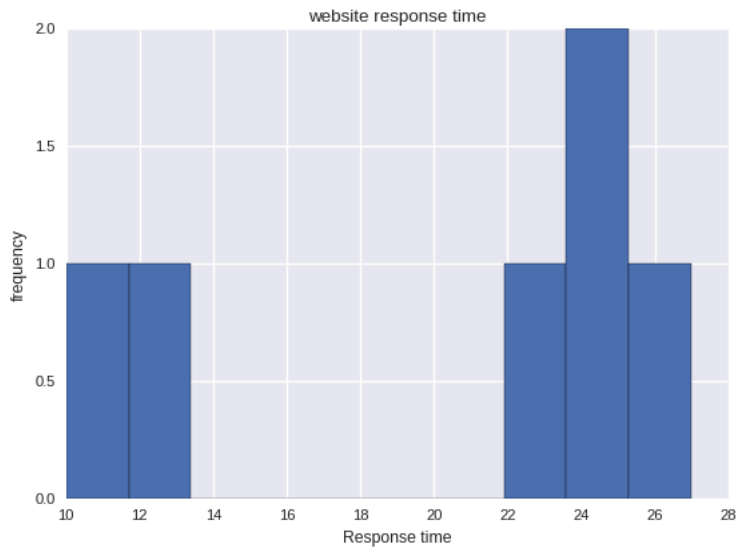



```
#Plot a histogram to analyze age distribution of people in a survey.

age=[10,18,19,17,20,31,40,50,60,66,70]
plt.hist(age,bins=10,color=["skyblue"])
plt.xlabel("Age")
plt.ylabel("frequency")
plt.title("Age distribution")
plt.show()
```

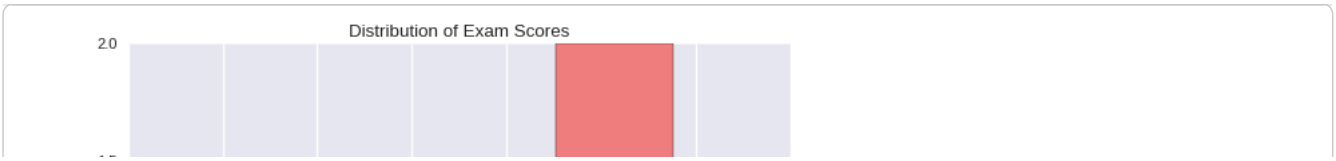
```
#Draw a histogram for salary distribution of employees in an organization.


salary=[5000,55000,340000,23000,100000,78000]
plt.hist(salary,bins=10,color=["skyblue"])
plt.xlabel("Salary")
plt.ylabel("frequency")
plt.title("Sslary distribution")
plt.show()
```




```
#Create a histogram to analyze website response times.

response_time_hr=[10,24,23,12,25,27]
plt.hist(response_time_hr,bins=10)
plt.xlabel("Response time")
plt.ylabel("frequency")
plt.title("website response time")
plt.show()
```




```
#Plot a histogram to understand how exam scores are spread across different ranges.

marks=[50,70,28,70,90]
plt.hist(marks, bins=5, color='lightcoral', edgecolor='black')
plt.xlabel("Exam Scores")
plt.ylabel("Frequency")
plt.title("Distribution of Exam Scores")
plt.grid(axis='y', alpha=0.75)
plt.show()
```


Distribution of Exam Scores
2.0